Levi Maia Braga

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Education

Centro Universitario Padre Anchieta — Jundiai, SP, Brazil
Bachelor of Science in Computer Science | GPA: 9.5/10.0 — *Aug. 2024 – Expected Dec. 2028*

- **Relevant coursework:** Data Structures, Object-Oriented Programming, Databases, Operating Systems and Embedded Systems, Cloud Computing, Systems Security, Artificial Intelligence, Calculus I–II

Projects

gpu-bench | *Python, CUDA, PyTorch, CuPy, NumPy, Linux* — Oct. 2026 – Present

- Built a GPU benchmark harness timing matrix multiply, SAXPY, reduction and host-to-device copies across NumPy, CuPy and PyTorch, with CUDA events, a synchronized CPU timer, and an unsynchronized timer kept as a control
- Measured 6.5 TFLOP/s on a 4096×4096 matmul on an RTX 2060 and 4.4 TFLOP/s on a Tesla T4, 38× NumPy on each machine's CPU, with every result checked against a float64 reference
- Quantified the cost of skipping synchronization: on the same matmul, an unsynchronized timer reported 0.21 ms (RTX 2060) and 0.073 ms (T4) against 21.0 and 31.2 ms with CUDA events, errors of 100× and 425×
- Ran an independent audit of code and docs against the raw CSVs, which found PyTorch's CUDA events being created inside the timed region; fixed it, re-measured both GPUs against predictions written in advance, and published the before and after
- Committed a written prediction before every run, and discarded one taken under desktop load, where NumPy, unchanged, took 29% longer; the rerun's written control (NumPy within 5%) passed at 1%
- Stamped every result with GPU, driver, CUDA and library versions, plus the git commit and seed since the audit; typed with mypy `--strict`, tested with pytest

SerpVive | *TypeScript, PostgreSQL, external APIs* — Mar. – Apr. 2026

- Built and deployed a production web application that ingests Google Search Console data through cron-scheduled pipelines and flags pages losing organic traffic, running unattended
- Designed a deterministic statistical scoring engine — decay over a 16-month window, 7/28-day velocity, seasonality check and a noise filter — and kept LLMs out of it, reserving model calls for the one step needing interpretation
- Engineered a four-provider failover chain across three external APIs with per-attempt timeouts, permanent-error classification and automatic degradation; both failover levels fired in production
- Modeled a 13-table PostgreSQL schema with row-level security on every table and 30 versioned migrations, specified before the first line of code and kept across 300+ commits

Technical Skills

Languages: Python, TypeScript, SQL (PostgreSQL), Bash
GPU and Performance: CUDA (events, host-device synchronization), PyTorch, CuPy, NumPy, benchmarking methodology (warm-up, median, relative standard deviation), GFLOP/s and effective bandwidth
Systems and Infrastructure: Linux, WSL2, shell scripting, REST APIs, OAuth 2.0, scheduled pipelines, multi-provider failover, schema design, versioned migrations, row-level security
Tools: Git, GitHub, uv, pytest, mypy, ruff, Next.js, React, Zod, Vercel, SSH

Awards

Gold Medal, Brazilian Olympiad of Astronomy and Astronautics (OBA) — 2021 and 2023
Honorable Mention, Brazilian National Science Olympiad (ONC) — 2022
Honorable Distinction, Kangaroo Mathematics Competition Brazil — 2022

Languages

English: Advanced; near-native comprehension from childhood in the United States, fluent technical writing
Portuguese: Native